ST11) H_0: $\xi \sim \rho_0(x) = \begin{cases} 1, & x \in (0;1) \\ 0, & x \notin (0;1) \end{cases} = 1\{(0;1)\}$

H_1: $\xi \sim \rho_1(x) = \begin{cases} \frac{e^{1-x}}{e-1}, & x \in (0;1) \\ 0, & x \notin (0;1) \end{cases} = \frac{e^{1-x}}{e-1}\{(0;1)\}$

a) $n = 1$, ур. знач. α

$$l(X) = \frac{L_1}{L_0} = \frac{p_1(x)}{p_0(x)} = \frac{e^{1-x}}{e-1} \geq C$$

$G: x \leq A$

$P(x \leq A \mid H_0) = \alpha$

$$\int_0^A 1\,dx = \alpha; \quad A = \alpha; \quad \underline{G: x \le \alpha}$$

$$W = P(x \le A \mid H_1) = \int_0^A \frac{e^{1-x}}{e-1}\,dx = \frac{-e}{e-1} \cdot e^{-x}\Big|_0^A =$$

$$= \frac{e}{e-1}\left(1 - e^{-\alpha}\right)$$

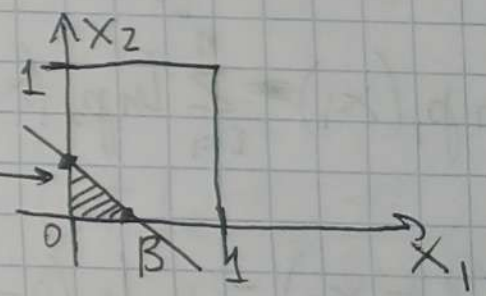
$$\alpha_1 = \alpha; \quad \alpha_2 = 1 - W$$

в) $n = 2$; ур. знач. α

$$l = \frac{L_1}{L_0} = \frac{\frac{e^{1-x_1}}{e-1} \cdot \frac{e^{1-x_2}}{e-1}}{1 \cdot 1} = \frac{e^{2-x_1-x_2}}{(e-1)^2} \ge e =$$

$$G: x_1 + x_2 \le B$$

$$P(x_1 + x_2 \le B \mid H_0) = \alpha$$



$$\iint 1 \cdot 1\,dx_1\,dx_2 = \alpha$$

$x_1 + x_2 \le B$

$0 \le x_1 \le 1$

$0 \le x_2 \le 1$

$$S = \frac{B \cdot B}{2} = \alpha \Rightarrow B = \sqrt{2\alpha}$$

$$B \le 1; \quad \sqrt{2\alpha} < 1; \quad \underline{\alpha < \frac{1}{2}}$$

$$G: x_1 + x_2 \le B = \sqrt{2\alpha}$$

$$W = P(x_1 + x_2 \le B \mid H_1) = \iint_{\substack{x_1 + x_2 \le B \\ 0 \le x_i \le 1}} \frac{e^{1-x_1} \cdot e^{1-x_2}}{(e-1)^2}\,dx_1\,dx_2 =$$

$$= \frac{e^2}{(e-1)^2} \int_0^B dx_1 \int_0^{B-x_1} e^{-x_1} e^{-x_2} dx_2 = \left(\frac{e}{e-1}\right)^2 \cdot$$

$$\cdot \int_0^B e^{-x_1}\left(1-e^{-B+x_1}\right)dx_1 = \left(\frac{e}{e-1}\right)^2 \left[1-e^{-B}-e^{-B}B\right]$$

$\alpha_1 = \alpha$

$\alpha_2 = 1 - W$

c) асимптот. крит. по выборке объема n; ур. знач. α

~~Нулевая~~ $$L(\vec{x}_n) = \frac{L_1(\vec{x}_n)}{L_0(\vec{x}_n)} = \frac{\prod_{i=1}^n p_1(x_i)}{\prod_{i=1}^n p_0(x_i)} = \prod_{i=1}^n \frac{p_1(x_i)}{p_0(x_i)} \geq C$$

$$\ln L = \sum_{i=1}^n \underbrace{\ln \frac{p_1(x_i)}{p_0(x_i)}}_{\eta_i} \geq \ln C$$

$$\sum_{i=1}^n \ln p_1(x_i) - \sum_{i=1}^n \ln p_0(x_i) = \sum_{i=1}^n \ln\left(\frac{e^{1-x}}{e-1}\right) - 0 \geq \ln C$$

$$\sum_{i=1}^n (1-x_i) - n\ln(e-1) \geq \ln C$$

$$\text{б: } \underbrace{\sum_{i=1}^n x_i \leq A}_{\bar{x} \leq B} \qquad \underbrace{P(\bar{x} \leq B \mid H_0) = \alpha}_{\text{ЦПТ (прост.):}}$$

$$\frac{\bar{x} - M\xi}{\sqrt{D\xi}}\sqrt{n} \sim N(0;1)$$

H_0: $M\xi = \int_0^1 x \cdot 1\,dx = \frac{1}{2}$; $M\xi^2 = \int_0^1 x^2 \cdot 1\,dx = \frac{1}{3}$

$D\xi = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$

$$P(\bar{x} \leq B \mid H_0) = P\Big(\frac{\bar{x} - \frac{1}{2}}{\sqrt{1/12}} \cdot \sqrt{n} \leq \underbrace{\frac{B - 1/2}{\sqrt{1/12}}\sqrt{n}}_{u_\alpha}\Big) = \alpha$$

$$B = \frac{1}{2} + \frac{u_\alpha}{\sqrt{12n}};$$

$$G: \bar{x} \leq \frac{1}{2} + \frac{u_\alpha}{\sqrt{12n}}$$

$$W = P(\bar{x} \leq B \mid H_1)$$

H_1: $M\xi = \int_0^1 x \cdot \frac{e^{1-x}}{e-1}\,dx = \frac{-e}{e-1}\int_0^1 x\,de^{-x} = \frac{-e}{e-1}\left(xe^{-x}\Big|_0^1 - \int_0^1 e^{-x}dx\right) =$

$= \frac{e}{e-1}\left(\frac{1}{e} + e^{-x}\Big|_0^1\right) = \frac{-1}{e-1} - \frac{e}{e-1}\cdot\left(\frac{1}{e} - 1\right) = \frac{-2}{e-1} + \frac{e}{e-1} =$

$= \frac{e-2}{e-1}$

$M\xi^2 = \int_0^1 x^2 \frac{e^{1-x}}{e-1}\,dx = \frac{-e}{e-1}\int_0^1 x^2\,de^{-x} = \frac{-e}{e-1}\left(x^2e^{-x}\Big|_0^1 - \right.$

$\left. -2\int_0^1 e^{-x}x\,dx\right) = \frac{-e}{e-1}\left(\frac{1}{e} - 2\cdot\int_0^1 e^{-x}x\,dx\right) =$

$= \frac{-1}{e-1} + 2\cdot\frac{e-2}{e-1} = \frac{2e-5}{e-1}$

$$D\xi = \frac{2e-5}{e-1} - \frac{e^2-4e+4}{(e-1)^2} = \frac{2e^2-2e-5e+5-e^2+4e-4}{(e-1)^2} =$$

$$= \frac{e^2-3e+1}{(e-1)^2}$$

$$P\left(\frac{\bar{x}-\mu_\xi}{\sqrt{D\xi}}\sqrt{n} \le \underbrace{\frac{B-\mu_\xi}{\sqrt{D\xi}}\sqrt{n}}_{\gamma}\right) = \int_{-\infty}^{\gamma} \frac{1}{2\pi} e^{-\frac{x^2}{2}} dx$$

<u>состоятельность критерия:</u> $W \xrightarrow[n\to\infty]{} 1$

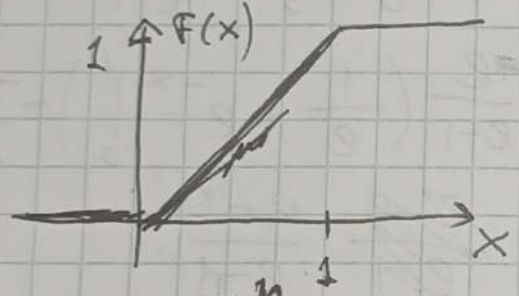
$\forall$ распред. из H_1

$$\gamma = \frac{\frac{1}{2} + \frac{u_\alpha}{\sqrt{12n}} - 0{,}418}{\sqrt{D\xi}}\sqrt{n} = \frac{\frac{u_\alpha}{\sqrt{12}} + 0{,}082\sqrt{n}}{\sqrt{D\xi}} \xrightarrow[n\to\infty]{} +\infty$$

<u>не сост.</u>

$\alpha_1 = \alpha$; $\alpha_2 = 1 - W$

d) $x_{min} < c$ – крит. область



$\alpha_2 \to \min$

$\alpha_1 \le \alpha$

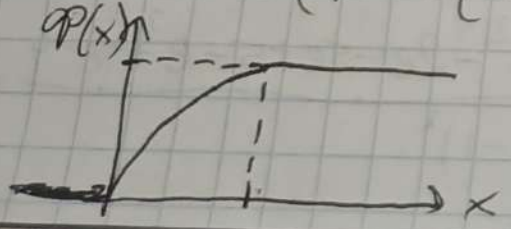
$$\alpha_1 = P(x_{min} < c \mid H_0) = 1 - (1 - F(c))^n \le \alpha$$

$$1 - (1-c)^n \le \alpha; \quad 1 - c \ge \sqrt[n]{1-\alpha};$$

$$c \le 1 - \sqrt[n]{1-\alpha}$$

$$\alpha_2 = P(x_{min} \ge c \mid H_1) = 1 - P(x_{min} < c \mid H_1) =$$

$$= \cancel{1} - \cancel{1} + (1 - \varphi(c))^n = (1 - \varphi(c))^n \longrightarrow \min$$



$$\cancel{\varphi(x) = \int_{-\infty}^{x} \frac{e^{t}}{e-1} dt = \frac{e}{e-1}\int_{-\infty}^{x} e^{-t} dt =}$$

$$\Phi(x) = \int_{-\infty}^{x} \varphi(x)\,dx = \int_{0}^{x} \frac{e^{1-t}}{e-1}\,dt = \frac{e}{e-1}\int_{0}^{x} e^{-t}\,dt =$$

$$= \frac{-e}{e-1}\cdot e^{-t}\Big|_{0}^{x} = \frac{-e}{e-1}\left(e^{-x}-1\right) = \frac{e}{e-1}\left(1-e^{-x}\right)$$

$$\frac{d\left(\left(1-\Phi(c)\right)^{n}\right)}{dc} = n\left(1-\Phi(c)\right)^{n-1}\cdot\left(-\Phi'(c)\right) =$$

$$= -n\left(1-\Phi(c)\right)^{n-1}\cdot\frac{+e}{e-1}\cdot e^{-c} = 0$$

$$1-\frac{e}{e-1}\left(1-e^{-c}\right) = 0; \quad 1-e^{-c} = \frac{e-1}{e}$$

$$e^{-c} = 1-\frac{e-1}{e} = \frac{1}{e} = e^{-1}; \quad \underline{c=1.}$$

$$\alpha_1 \le \alpha \Rightarrow 1-\sqrt[n]{1-\alpha} \geqslant c$$

$$0 \le 1-\alpha \le 1 \Rightarrow 0 \le \sqrt[n]{1-\alpha} \le 1;\ 0 \le 1-\sqrt[n]{1-\alpha} \le 1 \Rightarrow$$

$$\Rightarrow 0 \le c \le 1$$

$$\frac{d^2\left(\left(1-\Phi(c)\right)^{n}\right)}{dc^2} = +n(n-1)\left(1-\Phi(c)\right)^{n-2}\cdot\frac{e}{e-1}\cdot e^{-c} +$$

$$+n\left(1-\Phi(c)\right)^{n-1}\cdot\frac{e}{e-1}\cdot e^{-c} > 0$$

$$\Phi(c) < 1$$

$\Rightarrow$ $c=1$ — т. мин.

$$c \le 1-\sqrt[n]{1-\alpha} \Rightarrow \underline{c = 1-\sqrt[n]{1-\alpha}}$$

$\Rightarrow G: x_{min} < 1-\sqrt[n]{1-\alpha}$

$W = P(x_{min} < 1-\sqrt[n]{1-\alpha} \mid H_1) = 1-(1-\varphi(c))^n =$

$= 1-\left(1-\frac{e}{e-1}\left(1-e^{-c}\right)\right)^n; \quad c = 1-\sqrt[n]{1-\alpha}$

$W = 1-\left(1-\frac{e}{e-1}\left(1-e^{-1+\sqrt[n]{1-\alpha}}\right)\right)^n$

$\sqrt[n]{1-\alpha} = e^{\frac{1}{n}\ln(1-\alpha)} = 1+\frac{1}{n}\ln(1-\alpha)+o\left(\frac{1}{n}\right)$

$-1+\sqrt[n]{1-\alpha} = \frac{1}{n}\ln(1-\alpha)+o\left(\frac{1}{n}\right)$

$e^{-1+\sqrt[n]{1-\alpha}} = e^{\left(\frac{1}{n}\ln(1-\alpha)+o\left(\frac{1}{n}\right)\right)} = 1+\frac{1}{n}\ln(1-\alpha)+o\left(\frac{1}{n}\right)$

$W = 1-\left(1-\frac{e}{e-1}\left(\cancel{1}-\cancel{1}-\frac{1}{n}\ln(1-\alpha)+o\left(\frac{1}{n}\right)\right)\right)^n =$

$= \cancel{1-\left(1+\frac{e}{n(e-1)}\ln(1-\alpha)\right)}\ 1-\left(1+\frac{e}{n(e-1)}\ln(1-\alpha)+\right.$

$\left.+o\left(\frac{1}{n}\right)\right)^n \xrightarrow[n\to\infty]{} 1-e^{\frac{e\ln(1-\alpha)}{e-1}} = 1-(1-\alpha)^{\frac{e}{e-1}} \neq 1$

$\Rightarrow$ крит. не явл. состоят.

$\alpha_1 = \alpha; \quad \alpha_2 = 1-W$

5 T12) $P(4) = \frac{8}{24} = \frac{1}{3}; \quad P(3) = \frac{4}{24} = \frac{1}{6}$

$P(1) = P(2) = \frac{1}{4}$. — H_0

H_1: кость честная: $p(1)=p(2)=p(3)=p(4)=\frac{1}{4}$

$\alpha = 0{,}2; \quad n=2$

$$l = \frac{L_1(\vec{x}_n)}{L_0(\vec{x}_n)} \geq c \; ; \quad P(l \geq c \mid H_0) \leq \alpha$$

p_0 (при H_0):

	1	2	3	4
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{24}$	$\frac{1}{12}$
2	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{24}$	$\frac{1}{12}$
3	$\frac{1}{24}$	$\frac{1}{24}$	$\frac{1}{36}$	$\frac{1}{18}$
4	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{18}$	$\frac{1}{9}$

p_1 (при H_1):

	1	2	3	4
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
2	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
3	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
4	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$

l:

	1	2	3	4
1	1	1	$\frac{3}{2}$	$\frac{3}{4}$
2	1	1	$\frac{3}{2}$	$\frac{3}{4}$
3	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{9}{4}$	$\frac{9}{8}$
4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{9}{8}$	$\frac{9}{16}$

$$\frac{1}{36} + \cancel{4}^{1} \cdot \frac{1}{\cancel{24}_{6}} = \frac{1+6}{36} = \underbrace{\frac{7}{36}}_{} = 0{,}194 < \alpha$$

$$\frac{7}{36} + \cancel{2}^{1} \cdot \frac{1}{\cancel{18}_{9}} = \frac{7+4}{36} = \frac{11}{36} = 0{,}306 > \alpha$$

$$\Rightarrow l \geq \frac{3}{2} = c$$

$$\underbrace{G: \; l \geq \frac{3}{2}}; \quad \alpha_1 = \frac{7}{36} \; ;$$

$$W = P(l \geq c \mid H_1) = 5 \cdot \frac{1}{16}$$

$$\alpha_2 = 1 - W = 1 - \frac{5}{16} = \frac{11}{16}$$

ST13) $\cancel{\xi \sim N(\theta_1; \theta_2^2)}$ — длина

$\cancel{\eta \sim N(\theta_1; \hat{\theta}_2^2)}$ — ширина

~~Верхний Елинет: $S_{\theta_1} = 5,722$; $\hat{S}_{\theta_2} = 4,612$ $n = 139$~~

~~Европа: $S_{\theta_1} = 6,161$, $\hat{S}_{\theta_2} = 5,055$ $m = 1000$~~

~~$H_0: \theta_{2\,елин.}^2 = \theta_{2\,евр.}^2$ | $H_0: \hat{\theta}_{2\,елин.}^2 = \hat{\theta}_{2\,евр.}^2$~~

~~$H_1: \theta_{2\,елин.}^2 \neq \theta_{2\,евр.}^2$ | $H_1: \hat{\theta}_{2\,елин.}^2 \neq \hat{\theta}_{2\,евр.}^2$~~

$$\cancel{\frac{\bar{X} - \theta_1}{\theta_2} \sqrt{n} \sim N(0,1)}$$

$$\cancel{\frac{(n-1)S^2}{\theta_2^2} \sim \chi^2(n-1)} \qquad \cancel{\Delta = \frac{\bar{X} - \theta_1}{\theta_2} \sqrt{n}}$$

$$\cancel{\Delta = \frac{(n-1)S^2}{\theta_2^2} \sim \chi^2(n-1)}$$

Дано: $\vec{X}_n$

$\xi \sim N(\theta_1; a^2)$ – Елинет $\qquad H_0: a^2 = b^2$

$\eta \sim N(\theta_1; b^2)$ – Европа $\qquad H_1: a^2 \neq b^2$

$S_x^* = 5,722$; $\quad S_y^* = 6,161$ $\qquad \alpha = 0,05$

$n = 139 \qquad m = 1000$

$$\Delta = \frac{S_x^2/a^2}{S_y^2/b^2} \sim F(n-1; m-1)$$

$G: \tilde{\Delta} \geq F_{1-\frac{\alpha}{2}}$

$\tilde{\Delta} = \frac{S_x^2}{S_y^2}$ или $\tilde{\Delta} \leq F_{\frac{\alpha}{2}}$

$\cancel{C = F_{\frac{\alpha}{2}}(n-1; m-1) = F_{\alpha}(138, 999)}$

$\cancel{P(\Delta \le c \mid H_0) = \alpha}$ $\cancel{P(\Delta < F_{\alpha/2}(n-1, m-1)) + P(\Delta \ge F_{1-\alpha/2}(n-1, m-1)) = \alpha}$

$\cancel{\tilde{\Delta} = \frac{5{,}722^2}{5{,}721^2} \approx 0{,}8626}$, $\cancel{\tilde{\Delta} = \frac{5{,}722^2}{6{,}161^2} = 0{,}8626}$

$\cancel{p\text{-value} = 2P(\Delta \le \tilde{\Delta} \mid H_0) = 2\int_{-\infty}^{\tilde{\Delta}} q(t)\,dt \approx 2 \cdot 0{,}1365 =}$

$\cancel{= 0{,}273}$

$\cancel{\frac{\alpha}{2} = 0{,}025 < 0{,}273 < 1 - \frac{\alpha}{2} = 0{,}975}$

$\cancel{\Rightarrow}$ нет основания отвергнуть гипотезу H_0

<u>ширина:</u>

$\cancel{\xi \sim N(\theta_1; a^2)}$ — Англия

$\cancel{\eta \sim N(\varphi; b^2)}$ — Европа

$\cancel{F_{\frac{\alpha}{2}}(138{,}999) =}$

$P(\Delta < F_{\alpha/2}(n-1; m-1)) + P(\Delta \ge F_{1-\alpha/2}(n-1; m-1)) = \alpha$

$p\text{-value} = 2P(\Delta \ge \tilde{\Delta} \mid H_0) = 2\int_{\tilde{\Delta}}^{+\infty} q(t)\,dt \approx 2 \cdot 0{,}136 =$

$\cancel{\tilde{\Delta} = \frac{5{,}722^2}{6{,}161^2} = 0{,}8626}$, $\tilde{\Delta} = \frac{6{,}164^2}{5{,}722^2} = 1{,}1594$

$= 0{,}272$

$\frac{\alpha}{2} < p\text{-value} < 1 - \frac{\alpha}{2}$

нет основания отвергнуть H_0

$\cancel{W(\theta_2)=}\mathbb{P}(\Delta \geq c \mid H_1) = \mathbb{P}(\tilde{\Delta} \leq F_{\alpha/2} \mid H_1) + \mathbb{P}(\tilde{\Delta} \geq F_{1-\alpha/2} \mid H_1) =$

$$= \int_{F_{\alpha/2}/\theta_2}^{+\infty} q(t)\,dt + \int_0^{F_{1-\alpha/2}/\theta_2} q(t)\,dt$$

<u>для ширины:</u>

$\xi \sim N(\theta_1; a^2)$ – Елинет $\qquad H_0: a^2 = b^2$

$\eta \sim N(\varphi, b^2)$ – Европ $\qquad H_1: a^2 \neq b^2$

$S_x = 4,612; \quad S_y = 5,056 \qquad \Delta \sim F(n-1; m-1)$

$$\tilde{\Delta} = \frac{5,056^2}{4,612^2} \approx 1,202$$

$$p\text{-value} = 2\mathbb{P}(\Delta \geq \tilde{\Delta} \mid H_0) = 2\int_{\tilde{\Delta}}^{+\infty} q(t)\,dt \approx 2 \cdot 0,0859 =$$

$= 0,1718$

$\frac{\alpha}{2} < 0,1718 < 1 - \frac{\alpha}{2} \Rightarrow$ нет оснований отвергнуть H_0

5+14) $\xi \sim N(a; 2)$; $\eta \sim N(b; 1)$

$n = 3 \qquad m = 2$

$x = \{-1,11; -6,10; 2,42\}$; $y = \{-2,29; -2,91\}$

$H_0: a = b \qquad H_1: a > b$

$$\frac{\bar{x} - a}{\sqrt{2}}\sqrt{3} \sim N(0;1) \qquad \frac{\bar{y} - b}{\sqrt{1}}\sqrt{2} \sim N(0;1)$$

$\bar{x} - a \sim N(0; \frac{2^2}{3})$; $\bar{y} - b \sim N(0; \frac{1^2}{2})$

$\bar{x} - \bar{y} - (a - b) \sim N(0; \frac{2}{3} + \frac{1}{2})$

$$\Delta = \frac{\bar{x} - \bar{y} - (a - b)}{\sqrt{\frac{2}{3} + \frac{1}{2}}} \sim N(0; 1)$$

$G: \Delta \geq u_{1-\alpha}$ $(\Delta > 1{,}64485)$

$H_0: a = b \Rightarrow a - b = 0$

$$\tilde{\Delta} = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{7}{6}}} \sim N(0; 1)$$ $\tilde{\Delta} = 0{,}928906$

$\Rightarrow$ нет причин отвергнуть H_0

$\bar{x} - \bar{y} \sim N(a - b; \frac{7}{6})$ – при H_1 (не в G)

$$W = P(\tilde{\Delta} \in G \mid H_1) = P(\tilde{\Delta} \geq u_{1-\alpha}(\underbrace{a - b}_{\theta}; \frac{7}{6})$$